Demographic Signatures in Digital Payment Fraud: A Machine Learning Framework for India's UPI Ecosystem

Project Report

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Abstract

Unified Payments Interface (UPI) transactions in India are increasingly screened by machine learning models that assign a fraud score to every payment. A recurring governance question is whether such models draw their predictive power from who the customer is, from which institution and channel carry the payment, or from what the transaction does. This study examines that question on a public synthetic UPI dataset of 250,000 transactions with a 2.00 per cent fraud rate. Predictors are separated into three conceptually clean families, demographic and geographic attributes, institutional and channel attributes, and transactional behaviour, and each family is tested in isolation under a leakage-safe pipeline that applies one-hot encoding and Synthetic Minority Over-sampling for Nominal and Continuous features (SMOTENC) inside the training folds only. A gradient boosted tree model reaches a test ROC-AUC of 0.872 on the 80/20 test set using all families and 0.866 using behavioural features alone, whereas the demographic family (0.510) and the institutional family (0.492) perform at chance. A DeLong paired test finds no significant difference between the full and behaviour-only models (difference 0.001, p = 0.811), and a five-fold cross-validated uplift over a restricted amount-and-type baseline of 0.129 (95 per cent confidence interval 0.122 to 0.135) confirms the model learns genuine signal. The operating threshold is selected on a validation set and locked before the untouched test set is scored; probabilities are recalibrated by isotonic regression, improving the Brier score from 0.0212 to 0.0144. An exploratory fairness audit across sender age groups finds a uniform false-positive rate between 2.1 and 2.4 per cent. The central result is a null one: in this dataset there is no evidence of incremental predictive value from the tested demographic or institutional attributes once behaviour is accounted for. Because the dataset is synthetic and its label-generating process is undocumented, the finding is stated as a property of this dataset and its data-generating process rather than of India's UPI ecosystem at large, and any operational use of demographic attributes should require independent evidence on real transaction data.

1 Introduction

1.1 Background and motivation

The Unified Payments Interface (UPI) has become the dominant retail payment rail in India, and the volume of instant account-to-account payments it carries continues to grow

rapidly, a growth that has been accompanied by a matching rise in payment fraud and in machine learning systems built to detect it (Hajek et al.; 2023; Tingfei et al.; 2023). Because manual review cannot scale to this volume, providers assign a fraud probability to every transaction in real time. The design of those models raises a question that is both practical and ethical: when a model flags a UPI payment, is it responding to who the customer is, such as their age or location, to which bank, device or network mediates the payment, or to what the transaction itself does, such as its amount, timing and type? The distinction matters because conditioning authentication or friction on demographic attributes carries fairness and regulatory risk, and is defensible only if those attributes genuinely improve detection.

In brief. Fraud models now score every UPI payment, so it matters whether their signal comes from identity, infrastructure or behaviour; only the last is free of demographic-profiling risk.

1.2 Problem statement

Much applied work on payment fraud reports a single aggregate accuracy figure and does not isolate the contribution of demographic information from behavioural information. Where feature groups are compared, the grouping is frequently inconsistent, with channel or institutional attributes such as device type or bank folded into a demographic category. This blurs the very comparison that a fairness-aware design should make sharp. The present study addresses that gap by defining three conceptually distinct feature families and measuring the standalone and incremental predictive value of each.

In brief. The literature rarely separates demographic from institutional and behavioural signal cleanly; this study makes that separation the core of its design.

1.3 Research questions and hypotheses

The study is organised around three formal hypotheses, stated in advance of the analysis so that the experiments test them rather than being shaped by their outcomes.

- **H1 (full versus restricted baseline).** A model using all feature families achieves a higher test ROC-AUC than a restricted machine learning baseline that uses only transaction amount and type.
- **H2 (incremental value of demographic and institutional attributes).** Adding demographic and institutional attributes to behavioural features does not significantly improve discrimination, tested with the DeLong paired ROC test. The study expects to fail to reject this null.
- **H3 (error-rate parity).** The false-positive rate of the operational model is approximately uniform across sender age groups.

The comparison model in H1 is described throughout as a *restricted amount-and-type machine learning baseline*. It is not a production rule engine, and no claim is made that it represents current industry practice; it is a deliberately impoverished learner that isolates how much predictive value the remaining features add.

In brief. Three hypotheses are fixed before modelling: the full model beats a restricted baseline (H1), demographics and institution add nothing over behaviour (H2, a null the study expects to keep), and error rates are even across age groups (H3).

1.4 Scope and contribution

The analysis uses a public synthetic dataset, so its conclusions apply in the first instance to that dataset and its data-generating process; this limitation is stated prominently here and throughout rather than deferred to the closing pages. Within that scope the study contributes: a leakage-safe, reproducible pipeline with three cleanly separated feature families; a battery of statistical checks (DeLong test, cross-validated uplift with fold-level scores, a permutation control and a resampling-robustness comparison); a validation-selected and recalibrated operating point with a labelled confusion matrix; and an exploratory fairness audit reported through multiple metrics rather than a single rule.

In brief. The contribution is a rigorous, reproducible test of the demographic question, valid for this synthetic dataset and designed to be re-run on real UPI data.

2 Related Work

2.1 Machine learning for payment fraud

Supervised learning on historical transactions is the standard approach to card and mobile payment fraud. Gradient boosted trees are a common choice because they handle heterogeneous tabular features and non-linear interactions well; the XGBoost algorithm of Chen and Guestrin (2016) is widely adopted for this reason. In the mobile-payment setting, Hajek et al. (2023) propose an XGBoost-based framework that incorporates the financial cost of misclassification, and Tingfei et al. (2023) benchmark several learners for online payment fraud and document the difficulty created by extreme class imbalance. Feature engineering also matters: Bahnsen et al. (2016) and Whitrow et al. (2009) show that aggregating a transaction against recent customer history improves detection, motivating the treatment of behavioural context as a first-class signal.

In brief. Boosted trees are the established workhorse for tabular payment fraud, and prior work shows behavioural context is where detection gains come from.

2.2 Class imbalance

Fraud is rare, so classifiers trained on raw data tend to predict the majority class. The Synthetic Minority Over-sampling Technique of Chawla et al. (2002) addresses this by interpolating new minority examples, and the fifteen-year retrospective by Fernández et al. (2018) documents both its influence and its failure modes, including that naive interpolation between integer-coded categories produces meaningless synthetic points. This directly motivates SMOTENC, which resamples mixed categorical and numeric data correctly, and a comparison against native class weighting. López et al. (2013) caution that resampling interacts with data-intrinsic characteristics and should be validated rather than assumed beneficial, and Dal Pozzolo et al. (2018) emphasise realistic evaluation protocols for imbalanced fraud data.

In brief. SMOTE-style oversampling must respect categorical structure (hence SMOTENC) and must be validated against class weighting rather than assumed superior.

2.3 Evaluation under imbalance

At a two per cent base rate, accuracy is uninformative. Davis and Goadrich (2006) establish the formal relationship between precision-recall and ROC space, and Saito and Rehmsmeier (2015) show the precision-recall curve is more informative than ROC for imbalanced problems. Probability quality is assessed separately from ranking: Niculescu-Mizil and Caruana (2005) document how learners miscalibrate and how post-hoc methods correct them, and the Brier score of Good (1952) provides a proper scoring rule for calibrated probabilities.

In brief. Imbalanced fraud data demand PR-AUC alongside ROC-AUC, and probability calibration assessed separately from ranking accuracy.

2.4 Interpretability and fairness

The SHAP framework of Lundberg and Lee (2017) assigns each feature a theoretically grounded contribution to a prediction and is used here to check which families actually drive scores. On fairness, Hardt et al. (2016) formalise equal-opportunity and error-rate parity, while Chouldechova (2017) proves that calibration and error-rate balance cannot generally hold simultaneously when base rates differ across groups. This impossibility result is why the present audit reports both selection rates and error rates, and why Barocas and Selbst (2016) caution that a four-fifths selection-rate test alone cannot establish the absence of disparate impact.

In brief. SHAP provides the per-prediction lens, and fairness theory dictates reporting multiple, mutually incompatible metrics rather than any single rule.

2.5 Synthetic data in fraud research

Because genuine payment data are commercially sensitive and legally protected, synthetic datasets are widely used for method development in fraud research, and Dal Pozzolo et al. (2018) note that even released real datasets are transformed in ways that limit external validity. Synthetic data allow full reproducibility and safe publication, but any conclusion about feature importance inherits the assumptions of the generator. This study therefore treats the dataset as a controlled environment in which to demonstrate and validate a testing framework, and it reports the provenance limitation prominently rather than as a footnote.

In brief. Synthetic data enable open, reproducible fraud research but bound what any single study can claim; the framework, not the dataset, is what transfers.

2.6 Positioning of this study

The reviewed literature establishes the methods but rarely isolates the demographic contribution to fraud prediction under a clean feature-family design with a full statistical validation battery. Much UPI-specific published work appears in venues with limited peer review and is treated cautiously here; the present study grounds each methodological

choice in the peer-reviewed sources above and contributes a transparent, reproducible test of the demographic question on a clearly documented synthetic dataset.

In brief. This study fills the gap between strong general methods and weakly validated UPI-specific claims by testing the demographic question rigorously.

3 Data and Methodology

3.1 Dataset and provenance

The study uses the public *UPI Fraud Transactions Dataset* published on Kaggle (Mahendra; 2024): 250,000 transactions across one calendar year, fraud prevalence 2.00 per cent (5,008 of 250,000), imbalance approximately 1 to 48. Fields include a timestamp, transaction type and merchant category, amount, transaction status, sender and receiver age band, sender state, sender and receiver bank, device type and network type. The dataset is synthetic and its documentation does not describe how fraud labels were generated or whether demographic attributes were drawn independently of the label. This is stated plainly because it bears directly on interpretation: if demographic fields were generated independently of fraud, a null demographic result is partly a property of the data design. The finding is therefore framed throughout as conditional on this dataset.

In brief. A synthetic 250,000-row Kaggle dataset with a 2 per cent fraud rate; because its label-generation is undocumented, all conclusions are stated as conditional on it.

3.2 Scope decisions and leakage control

Two fields are excluded by design. The free-text `transaction_notes` field contains scam-message templates that effectively restate the label; a model retaining it reaches a near-perfect AUC, which is leakage rather than skill. The `transaction_status` field (success or failure) is excluded because its value is not known at the moment a payment is scored: a pre-transaction model must use only information available at scoring time. Its predictive content is in any case negligible, with fraud rates of 2.03 per cent on failed and 2.00 per cent on successful transactions. All remaining variables were reviewed for post-event information, and none encodes the outcome.

In brief. Notes are removed as label leakage and status as post-decision information, so every retained feature is available at scoring time.

3.3 Feature families

Predictors are grouped into three conceptually distinct families so the headline comparison isolates one question at a time: **demographic and geographic** (sender age band, receiver age band, sender state); **institutional and channel** (sender bank, receiver bank, device type, network type); and **behavioural and transactional** (transaction type, merchant category, amount, hour of day, day of week, weekend indicator). Keeping institutional and channel attributes separate from demographics is essential: banks, devices and networks describe the infrastructure of a payment, not the identity of a person, and conflating them would make any demographic conclusion unsound.

In brief. Three clean families, identity, infrastructure and behaviour, so the ablation answers exactly one question per family.

3.4 Encoding, resampling and pipeline discipline

All categorical variables are one-hot encoded, so no artificial ordering is imposed on nominal categories such as bank or state; this matters for the linear and neural models, where integer codes would fabricate a spurious ranking. Class imbalance is addressed with SMOTENC (Chawla et al.; 2002; Fernández et al.; 2018), designed for mixed categorical and numeric data, so no impossible fractional categories are created. A cost-weighted model using native class weighting (`scale_pos_weight`) is fitted as a robustness comparison. Every transformation, encoding, resampling and calibration, is fitted only on training data and, for cross-validation, refitted inside each fold.

In brief. One-hot encoding plus SMOTENC inside the training pipeline, with class weighting as a cross-check, guarantees no leakage and no invalid synthetic categories.

3.5 Models and evaluation protocols

The primary model is a gradient boosted tree classifier (Chen and Guestrin; 2016). Three further learners, logistic regression, a multilayer perceptron and a TensorFlow deep network, are trained on the identical pipeline as a cross-learner benchmark. Two evaluation protocols are used: Protocol A, an 80/20 stratified split for the family-isolation ablations, and Protocol B, a 70/15/15 stratified split whose validation partition selects the operating threshold before the test partition is scored. A chronological split (training on earlier months, testing on the final quarter) provides a temporal robustness check. Discrimination is measured by ROC-AUC and PR-AUC; the operating point by F1, precision, recall, a labelled confusion matrix and Precision@k; probability quality by Brier score and log-loss before and after calibration.

In brief. One primary model, three benchmark learners, two split protocols plus a temporal check, and an imbalance-aware metric battery.

4 Exploratory Analysis and Design

4.1 Fraud structure in the data

Before modelling, the data were examined descriptively. Figure 1 presents the temporal and demographic profiles side by side. Fraud rises in late-night hours, a behavioural pattern, while every demographic and institutional slice stays close to the 2.00 per cent base rate. A Welch t-test on the weekend indicator is not significant ($p = 0.131$), honest evidence that the calendar flag alone carries no signal. In contrast, fraud varies strongly by what the transaction does: from roughly half a per cent on bill payments to several times the base rate on peer-to-peer transfers, an almost eightfold behavioural swing, concentrated in specific type-by-hour combinations (Figure 2).

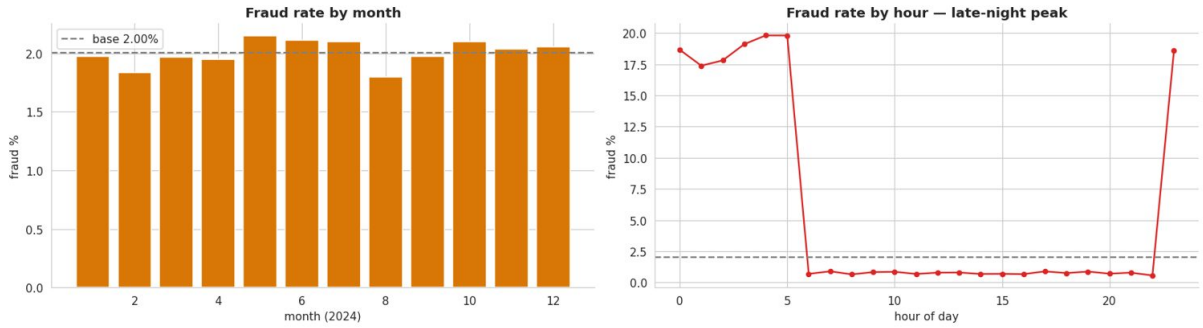
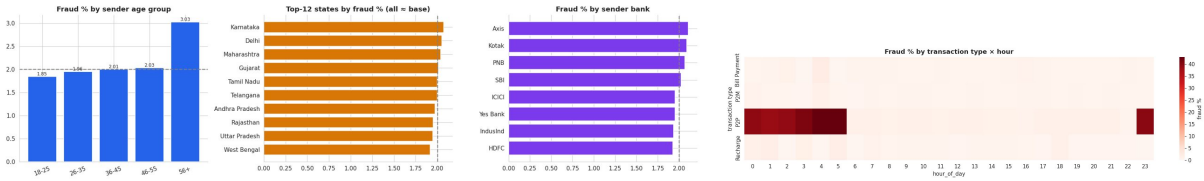


Figure 1: Fraud rate by calendar month and by hour of day. Fraud concentrates in late-night hours, a behavioural pattern.



(a) Age, state and bank: every slice near the base rate.

(b) Type × hour: risk concentrates behaviourally.

Figure 2: Demographic flatness against behavioural concentration.

In brief. Descriptively, fraud in this dataset is flat across identity and infrastructure but strongly structured by behaviour, foreshadowing the ablation.

4.2 System design

The end-to-end design is summarised in Figure 3, presented at full page so every stage is legible. Raw data pass through a scope-exclusion gate that removes the leakage-prone and post-event fields, then into the three-family feature design, the leakage-safe encoding and resampling pipeline, the two evaluation protocols, the modelling stage with its statistical and governance checks, and finally the reporting artefacts. The diagram mirrors the structure of the analysis notebook so each reported number can be traced to a stage.

In brief. The architecture is a single traceable flow from raw data to governed outputs, and it is reproduced at full-page scale for readability.

5 Implementation

5.1 Environment and reproducibility

The analysis is implemented in Python and runs end to end in a single seeded notebook on Google Colab. All stochastic components, the splits, the resampling and the model initialisation, are fixed to one random seed so the reported numbers reproduce on re-execution. The core libraries are scikit-learn (Pedregosa et al.; 2011), imbalanced-learn (Lemaître et al.; 2017) for SMOTENC, and XGBoost (Chen and Guestrin; 2016); the deep network uses TensorFlow. Exact pinned versions and reproduction steps are given in the configuration manual.

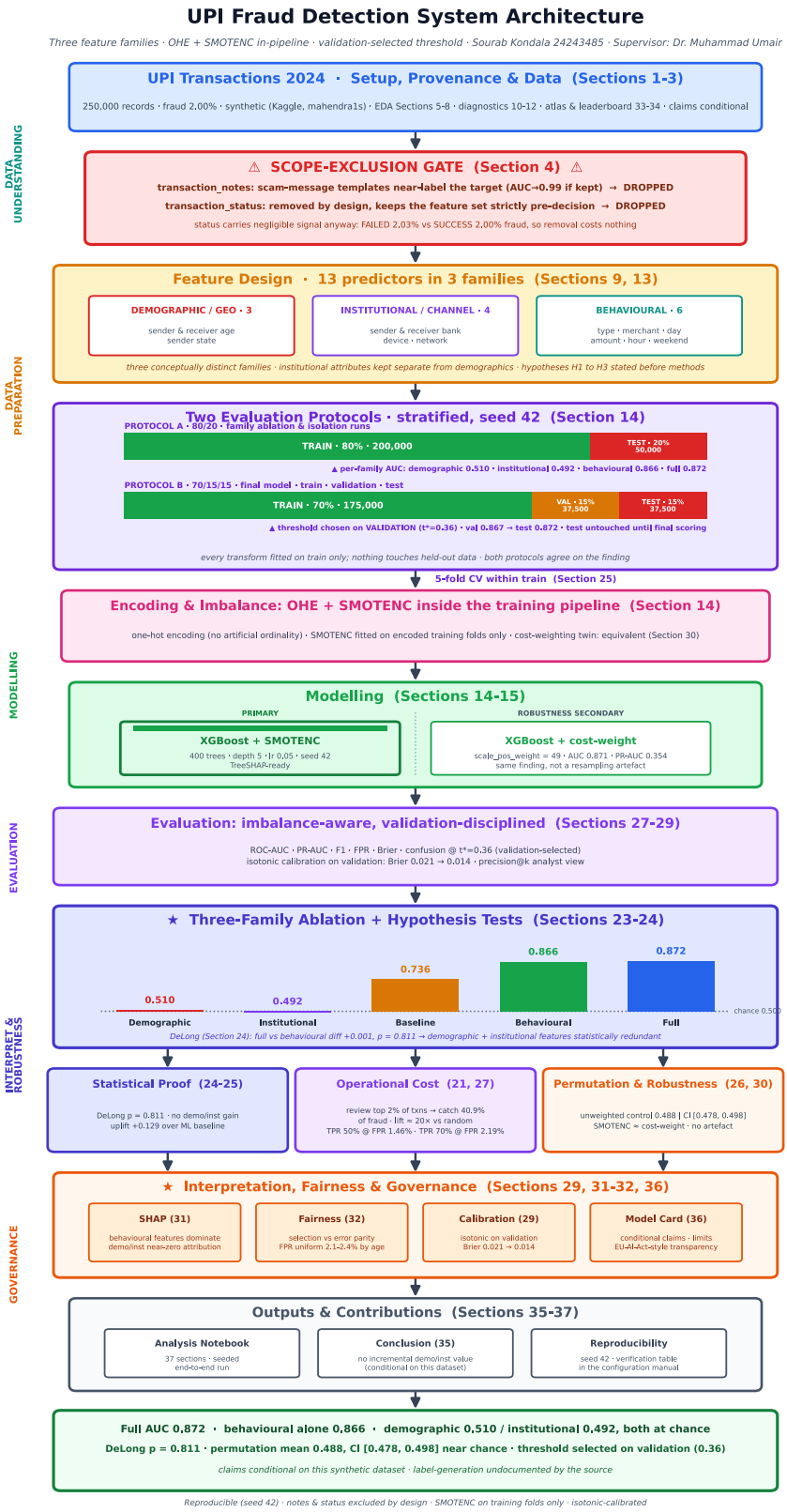


Figure 3: System architecture: from the scope-exclusion gate through the three feature families, the leakage-safe pipeline, the dual evaluation protocols, modelling with statistical and operational checks, to the governance and reporting layer.

In brief. One seeded notebook, pinned libraries, and a configuration manual that lets an examiner reproduce every number.

5.2 Pipeline construction and the family-isolation experiment

The training pipeline chains one-hot encoding, SMOTENC resampling and the classifier as a single estimator, so that inside cross-validation every transformation is refitted within each fold; at no point is the encoder or resampler exposed to validation or test rows. The primary classifier is a gradient boosted tree with a fixed configuration (400 trees, maximum depth 5, learning rate 0.05); the three comparison learners share the identical upstream pipeline. The central experiment retrains the identical model on one feature family at a time and records test ROC-AUC, so any performance difference is attributable to the family alone. The same procedure produces the restricted amount-and-type baseline for H1 and the behaviour-only model for H2.

In brief. Because the pipeline is one estimator refitted per fold, the family ablation isolates feature content, not preprocessing artefacts.

5.3 Statistical validation design

Three independent checks guard the headline result. A DeLong paired ROC test (DeLong et al.; 1988) compares the full and behaviour-only models on the same test transactions, accounting for their correlation. A five-fold cross-validated uplift over the baseline is computed with fold-level scores retained, its interval reported with the caveat that five folds give an indicative, not a large-sample, interval. A permutation control retrains the model on shuffled labels: a valid pipeline should then score near 0.50. A resampling-robustness comparison repeats the primary result with native class weighting instead of SMOTENC.

In brief. Four safeguards, DeLong, cross-validated uplift, a permutation control and a resampling swap, each target a different way the result could be spurious.

5.4 Operating point and calibration

The operating threshold is selected on the validation partition of Protocol B by maximising F1, then locked before the test partition is scored, so the test set is untouched by threshold choice as well as by training. Because oversampling inflates raw scores, probabilities are recalibrated by isotonic regression fitted on validation, and calibrated Brier score and log-loss are reported alongside raw values.

In brief. The threshold is chosen on validation and locked, and probabilities are recalibrated, so operational numbers on the test set are honest.

6 Evaluation

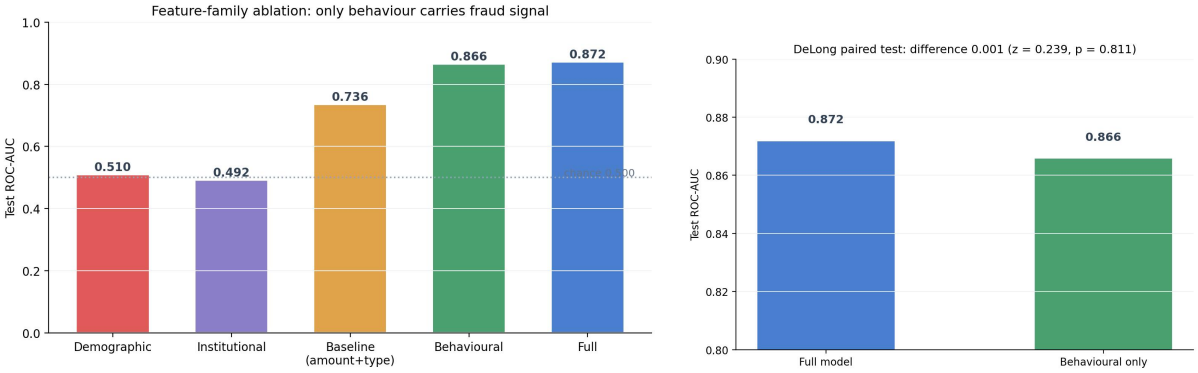
6.1 The central ablation and hypothesis tests

Table 1 and Figure 4(a) report the family-isolation experiment. Behavioural features alone reach a test ROC-AUC of 0.866 on the 80/20 test set, essentially equal to the full model at 0.872, while the demographic family (0.510) and the institutional family (0.492)

sit at the chance line. The restricted amount-and-type baseline reaches 0.736. **H1** is supported: a five-fold cross-validated uplift over the baseline gives a mean of 0.129 with a 95 per cent confidence interval of 0.122 to 0.135 (fold scores 0.124, 0.132, 0.129, 0.123, 0.134; paired t-test $p < 0.001$), an interval that is indicative rather than a large-sample guarantee. **H2** is not rejected: the DeLong paired test (DeLong et al.; 1988) of the full against the behaviour-only model gives a difference of 0.001 ($z = 0.239$, $p = 0.811$), shown in Figure 4(b). There is no evidence of incremental predictive value from the tested demographic or institutional attributes.

Feature family (trained in isolation)	**Test ROC-AUC**
Demographic / geographic	0.510
Institutional / channel	0.492
Behavioural	0.866
Restricted baseline (amount + type)	0.736
Full model (all families)	0.872

Table 1: Family-isolation ablation on the 80/20 test set (Protocol A).



(a) Ablation: only behaviour carries signal. **(b)** DeLong: difference 0.001, $p = 0.811$.

Figure 4: The central experiment and its paired significance test.

In brief. On the 80/20 test set, behaviour alone (0.866) matches the full model (0.872); demographics (0.510) and institution (0.492) sit at chance. H1 holds and H2 is kept, so the null demographic result is statistically grounded.

6.2 Validity and robustness checks

The permutation control retrains the model on shuffled labels and returns a mean ROC-AUC of 0.488 with a 95 per cent interval of 0.478 to 0.498 (Figure 5(a)), close to chance and evidence against label leakage; the small displacement below 0.500 is consistent with sampling variation under a single seed, and the check is treated as supportive rather than decisive. Replacing SMOTENC with native class weighting leaves ROC-AUC essentially unchanged (0.872 for the SMOTENC model versus 0.870 for the cost-weighted model, both on the 80/20 test set, Table 2) and slightly improves PR-AUC, so the finding is not an artefact of the oversampling method. The result also holds across evaluation protocols: 0.870 on the 80/20 test set, 0.870 on the 70/15/15 validation set and 0.873 on the 70/15/15 test set, and 0.867 on the chronological (temporal) test set training on earlier months and testing on the final quarter..

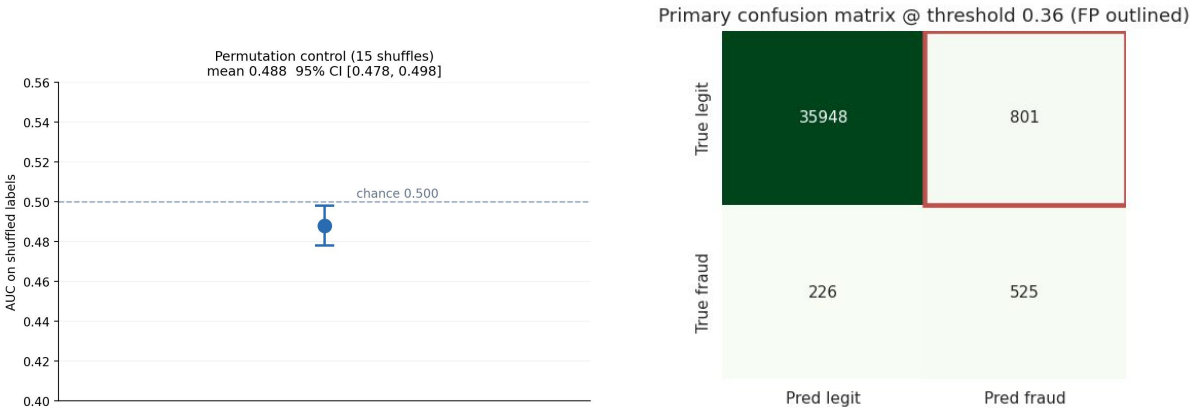
Imbalance method	ROC-AUC	PR-AUC
SMOTENC (primary)	0.872	0.329
Native class weighting (`scale_pos_weight`)	0.870	0.349

Table 2: Robustness to the imbalance method.

In brief. The result survives a shuffled-label control, a swap of the imbalance method, and three split protocols including a temporal one.

6.3 Operating point and calibration

The threshold selected on the validation partition is 0.36. Applied to the untouched test set it yields the confusion matrix in Figure 5(b): 525 true positives, 801 false positives, 226 false negatives and 35,948 true negatives, giving recall 0.70 at a false-positive rate of 2.2 per cent. Figure 6 presents the ROC and precision-recall curves, the latter the more informative at a two per cent base rate (Saito and Rehmsmeier; 2015). Reviewing only the highest-scoring two per cent of transactions captures roughly 41 per cent of all fraud, about twenty times the yield of random review. Isotonic recalibration on validation improves the primary model's raw Brier score from 0.0212 to 0.0144 and log-loss from 0.0996 to 0.0584 (Figure 8(a)); the 0.031 Brier in Table 3 is the raw, pre-calibration value reported for cross-model comparison under identical conditions.



(a) Permutation control near chance.

(b) Confusion at threshold 0.36.

Figure 5: Pipeline validity and the locked operating point.

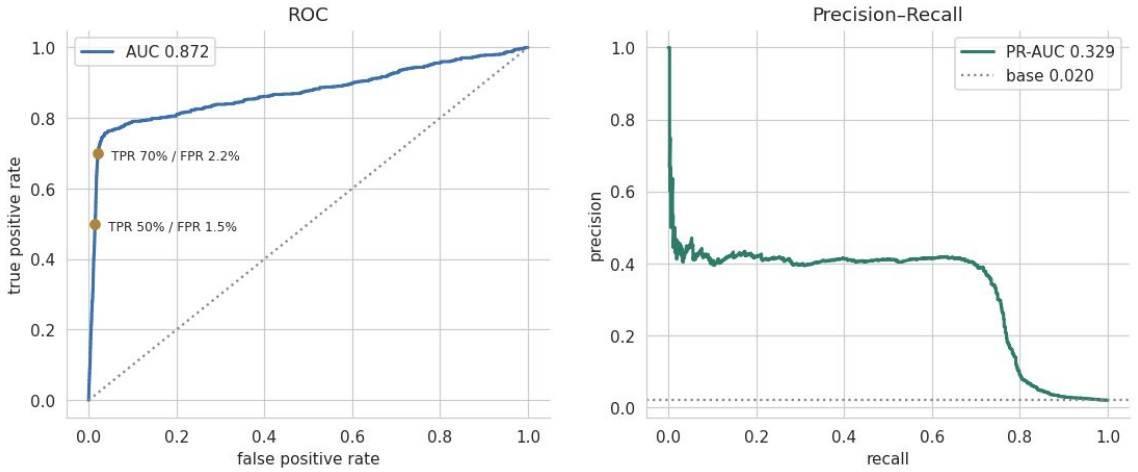


Figure 6: ROC (left) and precision-recall (right) curves for the full model on the test set.

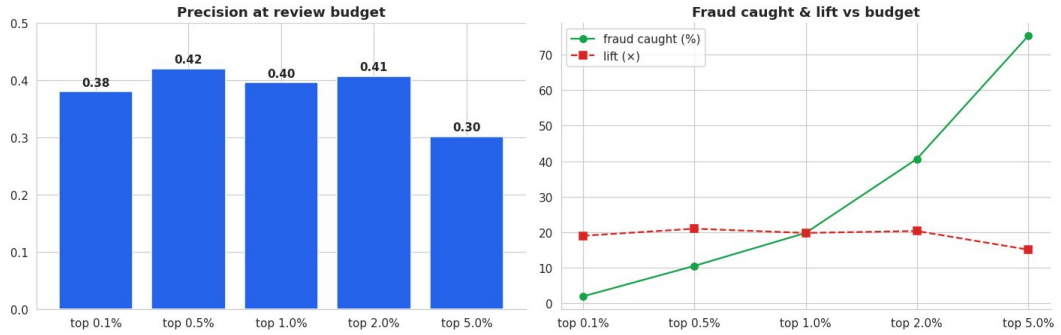


Figure 7: Precision@k analyst-budget view: reviewing the top two per cent of transactions by score captures about 41 per cent of all fraud.

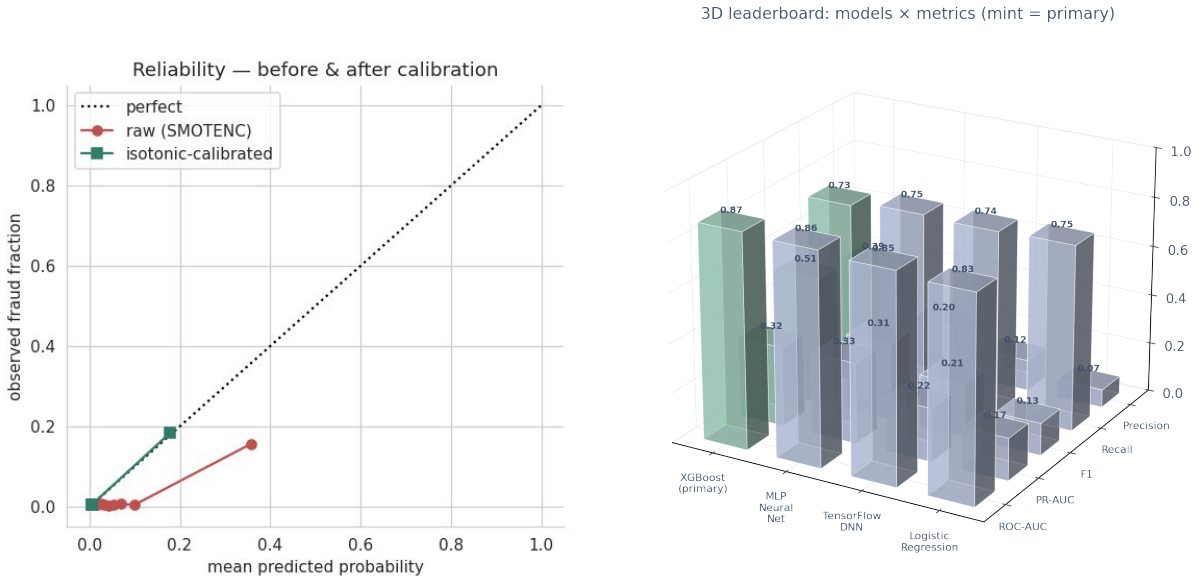
In brief. A validation-locked threshold gives recall 0.70 at 2.2 per cent FPR; a two per cent review budget catches 41 per cent of fraud; calibration makes the scores honest probabilities.

6.4 Cross-learner benchmark

Table 3 benchmarks four learners on the identical pipeline. The three non-linear models cluster together and the linear model trails, confirming the behavioural signal is not an artefact of one algorithm. The gradient boosted tree is retained as primary because it leads on ROC-AUC, holds the lowest false-positive rate, and supports exact tree-based SHAP explanations; Figure 8(b) presents the comparison across five metrics.

Model	ROC-AUC	PR-AUC	F1	Recall	Precision	FPR	Brier
XGBoost (primary)	0.870	0.321	0.509	0.729	0.392	0.023	0.031
MLP neural net	0.862	0.327	0.310	0.751	0.195	0.063	0.059
TensorFlow DNN	0.848	0.217	0.213	0.744	0.124	0.107	0.121
Logistic regression	0.830	0.170	0.130	0.751	0.071	0.201	0.145

Table 3: Cross-learner leaderboard on the 80/20 test set (Protocol A), sorted by ROC-AUC (Brier pre-calibration).



(a) Reliability before and after isotonic calibration.

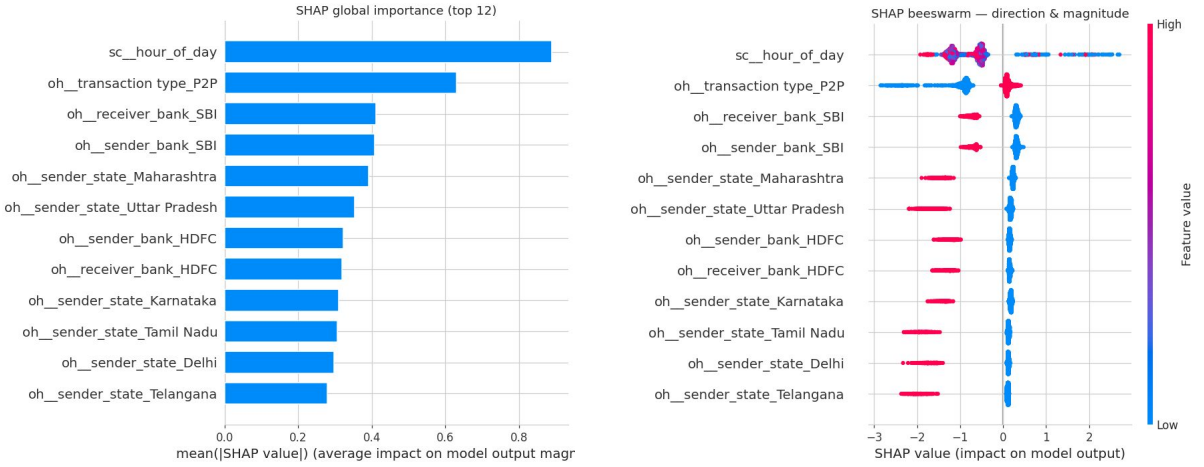
(b) Leaderboard across five metrics (mint = primary).

Figure 8: Probability quality and the cross-learner comparison.

In brief. Four learners on one pipeline tell the same story; the boosted tree stays primary for accuracy, low customer friction and exact explainability.

6.5 Explainability

Figure 9 shows global SHAP importance and the direction of effects (Lundberg and Lee; 2017). Behavioural features dominate; demographic and institutional features receive near-zero attribution, the individual-level echo of the ablation. Stated precisely: age is included in the full model but receives low attribution and does not improve discrimination in this dataset. It is not claimed that the model ignores age.



(a) Global importance.

(b) Direction and magnitude per prediction.

Figure 9: SHAP analysis: behavioural features dominate attribution.

In brief. SHAP confirms at the level of individual predictions what the ablation shows in aggregate: the model's decisions run through behaviour.

6.6 Fairness audit

The audit is exploratory and reported through multiple metrics by sender age group (Table 4, Figure 10), following Chouldechova (2017) on the impossibility of satisfying all criteria at once and Barocas and Selbst (2016) on the insufficiency of the four-fifths rule alone. Selection rate rises for the oldest group (ratio 0.708 against the least-flagged group, below the four-fifths mark), but the table shows why: that group's base fraud rate is higher (3.01 against roughly 1.9 per cent), so the model tracks outcome rates, not identity. The false-positive rate is uniform between 2.1 and 2.4 per cent, supporting **H3**. Within-group counts are modest and the data synthetic, so this supports but does not prove the absence of disparate impact, and no conclusion is drawn beyond the tested groups.

Age group	n	Base %	Selection	FPR	TPR	Precision
18–25	9,250	1.86	0.034	0.021	0.715	0.394
26–35	13,147	1.95	0.034	0.021	0.685	0.392
36–45	9,445	2.13	0.037	0.024	0.667	0.380
46–55	3,766	1.70	0.033	0.021	0.734	0.379
56+	1,892	3.01	0.047	0.023	0.790	0.511

Table 4: Exploratory fairness metrics by sender age group.

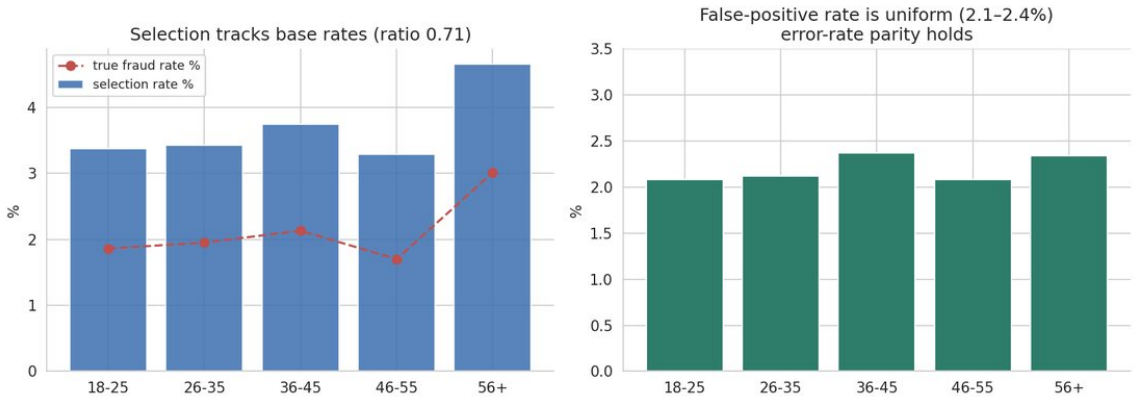


Figure 10: Fairness audit under two lenses: selection rate against base rate (left) and error-rate parity (right).

In brief. Selection differences mirror base-rate differences while the false-positive burden is even across ages (H3 holds); the audit is honest about what it can and cannot prove.

6.7 Summary of hypothesis outcomes

Table 5 consolidates the three pre-specified hypotheses, the tests applied and their outcomes, so the chain from question to evidence can be audited at a glance.

Hypothesis	Test	Outcome
H1: full model beats the restricted amount-and-type baseline	Five-fold cross-validated uplift with paired t-test	Supported: mean uplift 0.129, 95% CI [0.122, 0.135], $p < 0.001$
H2: demographic and institutional attributes add no discrimination over behaviour	DeLong paired ROC test, full vs behaviour-only	Not rejected: difference 0.001, $z = 0.239$, $p = 0.811$
H3: false-positive rate uniform across age groups	Group-wise FPR at the locked threshold	Supported: FPR between 2.1% and 2.4% for every group

Table 5: Pre-specified hypotheses, tests and outcomes.

In brief. All three hypotheses resolve as designed: H1 supported, H2 retained as the central null, H3 supported.

6.8 Threats to validity

Internal validity. The chief internal threat is information leakage, addressed by construction: the pipeline refits every transformation inside each fold, the threshold is chosen on validation, and the permutation control confirms the pipeline scores near chance on noise. **Construct validity.** The feature families operationalise the constructs of identity, infrastructure and behaviour; misassignment would blur the comparison, which is why banks, devices and networks are deliberately kept out of the demographic family. **External validity.** This is the binding constraint: the dataset is synthetic, so the null demographic finding may not transfer to real UPI traffic, and no such transfer is claimed. **Statistical conclusion validity.** The DeLong test accounts for the correlation between paired models, the cross-validated interval is flagged as five-fold and indicative, and the fairness audit is labelled exploratory because of small subgroup counts.

In brief. Each classic validity threat is either controlled by design (leakage, construct assignment) or disclosed as a boundary (external transfer, small subgroups).

7 Discussion and Conclusion

7.1 Interpretation

The evidence in this dataset is consistent and points one way. A model using every feature family reaches a test ROC-AUC of 0.872 on the 80/20 test set, a model using behaviour alone reaches 0.866 on the same split, and the difference is not statistically significant. Trained in isolation, the demographic and institutional families perform at chance. The result is not that demographic attributes are weak predictors but that, once behaviour is accounted for, they add no measurable discrimination. A behaviour-led fraud model can therefore match a full model while avoiding the fairness and regulatory exposure that comes with conditioning on customer identity.

In brief. The finding is an equivalence, not a ranking: behaviour alone does everything the full model does, so identity attributes buy nothing but risk.

7.2 Contribution

The contribution is a rigorously validated null result together with the framework that establishes it: three cleanly separated feature families, a leakage-safe pipeline, hypotheses fixed in advance, and a validation battery that rules out the most common ways such a result could be spurious. Rather than claiming novelty for the dataset or domain, the study offers a transparent, reproducible method for asking the demographic question, ready to be re-run on real transaction data.

In brief. A defended null result plus a reusable testing framework is the contribution, not a claim of first discovery.

7.3 Limitations

The dataset is synthetic and its label-generating process undocumented; if demographic attributes were generated independently of the label, the null result is partly a property of the data design, so the finding is conditional on this dataset rather than a claim about India's UPI ecosystem. The dataset contains no rolling velocity or account-history features, which would likely widen, not narrow, the reported gap. The fairness audit is exploratory with modest group counts. There is no real-time or concept-drift evaluation. The model is therefore not described as deployment-ready, and no claim is made that it provides direct guidance for live UPI authentication.

In brief. Synthetic data, no velocity features, small fairness subgroups and no drift testing bound the claim; none of these is hidden.

7.4 Future work

The natural next step is to repeat the identical framework on a real, well-documented UPI dataset with behavioural velocity features and larger demographic subgroups, adding drift monitoring and a recalibration schedule. The permutation control would benefit from a larger number of shuffles to tighten its interval onto 0.50. If the demographic families remain at chance under those richer conditions, the null result would carry beyond the synthetic setting.

In brief. Re-run the framework on real data with velocity features and more shuffles; the design transfers unchanged.

7.5 Ethics and responsible use

The study touches personal attributes, age and location, so its governance posture matters as much as its accuracy. Three practices follow from the findings and from the fairness literature (Hardt et al.; 2016; Chouldechova; 2017): fraud models should be accompanied by a model card that states their scope and known limitations; demographic attributes should be excluded from scoring unless their incremental value is demonstrated on real data under audit; and fairness reporting should present several metrics together, since no single criterion can certify equitable treatment. The framework developed here produces

all three artefacts as standard outputs, which is intended as a template for transparent deployment review rather than as evidence of deployment readiness.

In brief. The responsible path is behaviour-led scoring, a published model card and multi-metric fairness reporting; this framework generates all three by default.

7.6 Conclusion

Framed as three feature families and tested with hypotheses fixed in advance, UPI fraud in this synthetic dataset is a behavioural phenomenon. There is no evidence of incremental predictive value from the tested demographic or institutional attributes once behaviour is accounted for, and a behaviour-led model achieves a test ROC-AUC of 0.872 on the 80/20 test set while treating legitimate customers evenly across age groups. Whether that pattern holds on real data is an empirical question this framework is designed to answer, and any operational use of demographic attributes should await that evidence.

In brief. Behaviour carries the signal in this dataset; demographic profiling adds nothing here, and its real-world use should require real-world evidence.

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